

Project Manager Maneuver Ammunition Systems
Product Lead Special Ammunition and Weapon Systems

GENERAL SPECIFICATION FOR
SMALL CALIBER AMMUNITION
10 February 2026

1. SCOPE

1.1 Scope. This general specification describes the performance requirements and verification procedures for Small Caliber Ammunition.

2. APPLICABLE DOCUMENTS:

2.1 General. In addition to documents listed in this Section, there are documents specified in Section 5 of this specification. All documents listed in this specification should use the latest revision available at the time of award.

2.2 Government documents. Not Applicable

2.3 Non-Government publications.

- a) United Nations Recommendations on the Transport of Dangerous Goods Model Regulations, 24th Edition, 2025.
- b) International Civil Aviation Organization (ICAO) Technical Instructions for the Safe Transport of Dangerous Goods by Air, 2025-2026 Edition.
- c) International Maritime Organization (IMO) International Maritime Dangerous Goods Code (IMDG) 2024 Edition, Amendment 42-24.
- d) Sporting Arms & Ammunition Manufacturer's' Institute (SAAMI) Technical Committee Manual Volume III Centerfire Rifle, 2023
- e) Commission internationale permanente pour l'épreuve des armes à feu portatives (C.I.P.), <http://www.cip-bobp.org/en/home> (website shall be verified prior to TDP submission provided link is no longer functional)

2.4 Acquisition requirements. Acquisition documentation (base award and/or Request for Proposal (RFP)) must specify the following:

- a. Title and date of this specification
- b. Requirements for certificates of conformance for each lot or shipment of product.
- c. Requirements for age of ammunition.
- d. Requirements and provisions for submission of data as required.
- e. Requirements for acceptance criteria if different than those stated in specification.
- f. Requirements for reduction or elimination of verification procedures.
- g. Requirements and provisions for contractor and Government verification.
- h. Requirements and provisions for packaging of ammunition.
- i. Requirements and provisions for transportation of ammunition.

2.5 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Requirements specified in the contract or delivery order precede the text of this document. Nothing in this document or the contract, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Technical Requirements. Ammunition shall meet the requirements and verifications of this specification for the small caliber ammunition listed in ANNEX A. Any ammunition that is categorized as small caliber but is not specifically identified in ANNEX A must still adhere to the testing parameters of this specification. All small caliber ammunition shall be accompanied by technical data (see APPENDIX) and meet or exceed the requirements outlined in this specification.

3.1.1 Compliance to technical requirements and storage history. A Technical Data Package (TDP) as well as lot history folders shall be provided to ensure ammunition is in compliance with this specification. Additionally, documented storage history of the ammunition shall be provided.

3.2 Conformance Testing and Inspection. A random sample of the ammunition/items from each lot presented shall be selected by a U.S. Government (USG) Representative and subjected to conformance testing and inspection in accordance with TABLE I (to include retest rounds, if required).

3.2.1 Stock Ammunition. All ammunition shall be less than 10 years old and meet the verification procedures in TABLE I of this specification. Sufficient evidence of compliance to technical requirements and acceptable storage history of ammunition as determined by SFAE-AE-MAS-SAWS may warrant reduction or elimination of the verification procedures. Reduction of verification procedures does not apply to new production ammunition.

3.3 Serviceability. Ammunition shall be serviceable/issuable, in good condition and without visible signs of degradation.

3.3.1 Identification of Defects. Visual defects shall be identified within the technical data and include a defined method for acceptance/rejection of the ammunition.

3.3.2 Tip identification. Ammunition for which the manufacturer has specified, shall be permanently marked with tip identification.

3.3.3 Head stamp. Industry standard cartridge case head stamp is required on each cartridge of ammunition. At a minimum, the head stamp shall include year and code of the manufacturer.

3.4 Interface and interoperability with Weapon. The ammunition configuration shall be functionally compatible with the weapons identified in ANNEX A or as otherwise specified in contract or delivery order.

3.5 Operating requirements. All ammunition shall provide their functional, operational, and performance capabilities when fired from the weapons listed in ANNEX A regardless of weather related conditions (i.e. rain, snow, mud, wind, fog). If testing occurs in inclement weather, the Contractor shall implement all necessary safety precautions.

3.5.1 Operating temperatures. The cartridges must be capable of functioning reliably throughout a temperature range which includes ambient and extreme temperatures. All cartridges are required to reliably perform within a temperature range of -30°C to +50°C.

3.5.2 Velocity. The ammunition shall meet the average velocity requirement, at ambient temperature, as specified in ANNEX B.

3.5.3 Chamber pressure. The ammunition shall meet the individual and average chamber pressure requirements, at ambient temperature, as specified in ANNEX B.

3.5.4 Accuracy/Dispersion. The ammunition shall meet accuracy and/or dispersion requirements, at ambient, as specified in ANNEX B.

3.5.5 Function and casualty. The ammunition shall function without casualty or damage to weapon while maintaining metal parts security. Ammunition shall exhibit integrity during weapon usage and in ballistic flight as specified in Table II.

3.5.6 Bullet integrity. The jacket of the bullet, or any part thereof, shall not strip from the slug when the cartridge is fired.

3.5.7 Performance. The ammunition shall meet the intended bullet performance to include but not limited to trace function, incendiary performance, blank, proximity, airburst, anti-armor capability, etc. as detailed in the manufacturer's specifications. Ammunition shall be non-corrosive and free of mercury for U.S. customers.

3.6 Environmental Requirements.

3.6.1 Sequential rough handling. Ammunition shall be capable of withstanding the rigors of the sequential rough handling and transportation throughout extreme temperature ranges and meet all performance and safety requirements.

3.6.2 Shelf life. The ammunition shall meet all requirements of this specification within its operating temperature range at a minimum of ten (10) years. All cartridges are required to reliably be stored within a temperature range of -30°C to +50°C at a minimum of ten (10) years.

4. VERIFICATION

4.1 Technical verification. The ammunition shall be subject to all requirements of applicable technical data in addition to verification in accordance with TABLE I of this specification. Non-compliance to any requirements shall be cause to withhold acceptance of the lot or batch in which the non-compliance was found.

TABLE I. Requirements/Conformance Verification [Lot Acceptance Testing (LAT)]

Section 3 Requirements	Requirement	Section 4 Verification Procedures	Sample Size
3.1	Technical requirements	4.1	-
3.3	Serviceability	4.3	Note 1
3.3.1	Identification of defects	4.3	Note 1
3.3.2	Tip identification	4.3	Note 1
3.3.3	Head stamp	4.3	Note 1
3.4	Weapon interface and interoperability	4.4	ANNEX A
3.5.2	Velocity	4.5.2	ANNEX B
3.5.3	Chamber pressure	4.5.3	ANNEX B
3.5.4	Accuracy/Dispersion	4.5.4	ANNEX B
3.5.5	Function and casualty	4.5.5	ANNEX A
3.5.6	Bullet integrity	4.5.6	ANNEX A
-	Packing	-	Note 1
-	Marking	-	Note 1
Note 1: To be performed using a defined sampling method for inspection determined by the ammunition lot/batch size.			

4.1.1. Lot formation. The ammunition shall be assembled into identifiable lots, sublots, or batches, or in such other manner contingent on approval by USG. Each lot or batch shall, as far as practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time. Lot quantity shall be determined by the total quantity for shipment and include the additional cartridges from that same lot to be consumed during lot acceptance testing along with contingency cartridges for potential retesting. The lots or batches shall be identified by the contractor and shall be kept intact in adequate and suitable storage space. All cartridges in a given lot shall be manufactured prior to LAT testing.

4.1.1.1. Lot Size. All lot sizes are to remain consistent throughout an awarded delivery order. Once temperature testing is conducted, all skip lot sizes shall be within 20% of that lot's size. If a lot size larger than 20% is proposed, temperature testing has to be conducted on that lot. If a lot fails for any reason, the next two lots presented for additional testing have to be within 20% of the failed lot size.

4.1.1.2 Component Lots. If multiple lots of a component are present, a minimum of 10 rounds shall be fired for each lot.

4.1.2 Retest criteria. A ballistic retest is authorized upon initial failure of ammunition to meet test requirements as long as the failure is not considered critical. Retest procedures must be clearly defined in the technical data and results must be combined with data from the first test unless a failure analysis confirms that the failure was caused by something other than the ammunition (i.e. weapon failure or faulty test set up).

4.2 Verification of compliance. Evidence shall include, but is not limited to, the following:

- a. Identification of technical requirements to which ammunition is/was produced.
- b. Manufacturer, date of manufacture and original acceptance documentation.
- c. Initial acceptance reports, including type, lot or batch identification, quantity and method of acceptance (example: sample size, verification method, acceptance criteria, and results).
- d. Surveillance reports, including lot or batch identification, quantity and method of surveillance (example: sample size, verification method, criteria for action to be taken on lot or batch, results).
- e. Storage history, including duration and storage condition (example: controlled, uncontrolled).

4.3 Serviceability. A random sample shall be selected from each lot of ammunition for inspection. The ammunition shall be visually inspected in accordance with ANNEX C of this specification. Reinspection of a lot that fails for non-ballistic reasons is authorized as long as the criteria for reinspection is clearly outlined in the technical data.

4.4 Weapon Interface and interoperability verification. A random sample shall be selected from each lot of ammunition to be delivered and inserted into a weapon chamber or chamber gage that conforms to the identified weapon system to check for profile and alignment. Inability to interface properly with the weapon system shall be considered a failure to show compliance and constitute a single reliability failure.

4.5 Operating verification.

4.5.1 Operating temperatures. Cartridges shall operate reliably and safely when conditioned for not less than four (4) hours and fired throughout the extreme operating temperature range within five (5) minutes of removal from conditioning box or thermally insulated bag. Temperature testing shall be conducted during Function & Casualty tests and on a skip-lot basis (see APPENDIX) during LAT. Functional testing parameters must be defined in the technical data. Cartridges shall consistently perform within a temperature range of -30°C to +50°C, ensuring reliable function under the specified environmental conditions.

4.5.2 Velocity. Velocity tests shall be conducted in accordance with producer's test procedure to verify that a sample representative of the lot meets the specified velocity requirements in ANNEX B (10 cartridges minimum). The length of the test barrel must be representative of the weapon system in ANNEX A. If the lot fails to meet the Average Velocity or Δ Velocity requirement a retest

of double the sample size (20 cartridges minimum) is permitted. The cumulative average of all cartridges tested (30 cartridges minimum) shall be used to determine if the lot passes.

4.5.3 Chamber pressure. Chamber pressure tests shall be conducted during the LAT in accordance with the producer's test procedure to verify that a sample representative of the lot does not exceed the average or individual pressure requirements listed in ANNEX B (10 cartridges minimum). If the lot fails to meet the Average Chamber Pressure requirement a retest of double the sample size (20 cartridges minimum) is permitted. The cumulative average of all cartridges tested (30 cartridges minimum) shall be used to determine if the lot passes. Any single cartridge that exceeds the maximum individual pressure requirement will result in lot rejection; there shall be no retest.

4.5.4 Accuracy/dispersion. Accuracy/dispersion tests shall be conducted to verify that a representative sample of the lot meets the requirements identified in ANNEX B (10 cartridges minimum). Dispersion testing may be conducted at a distance other than those defined in ANNEX B with a proportional adjustment to mean radius versus range. If the lot fails to meet the Accuracy requirement or a bullet misses the target, a retest of double the sample size (20 cartridges minimum) is permitted. The cumulative average of all cartridges tested (30 cartridges minimum) shall be used to determine if the lot passes. Any additional bullet misses on target during retest shall result in lot rejection; there shall be no retest.

4.5.5 Function and casualty. A random sample shall be selected from the lot of ammunition to be delivered and functioned from the identified weapon system at all operating temperatures (extreme hot, cold and ambient) to check for function, casualty and metal parts security. Inability to function without casualty or damage to weapon while maintaining metal parts security shall be considered a failure and constitute a single reliability failure. See Table II for minimum function, casualty and metal parts security requirements.

TABLE II. Small Caliber Ammunition Function and Casualty Requirements

Criteria	Requirement (sample size - accept - reject) ¹	Retest
Any cartridge that either causes injury to user, damages the weapon or renders it inoperable (without the use of special tools) will be considered a critical defect.	230-0-1	Critical, No Retest
Bullet in bore	230-0-1	Critical, No Retest
Hangfire	230-0-1	Critical, No Retest
Firing without depression of trigger (self-firing)	230-0-1	Critical, No Retest
Misfire: primer does not function during first firing pin impact	230-1-2	350-2-3 (cumulative)
Failure to Chamber in weapon	230-1-2	350-2-3 (cumulative)
Failure to Extract	230-1-2	350-2-3 (cumulative)
For Blank cartridges only; Combined misfires, Failure to Feeds and Failure to Extracts	100-2-3	200-4-5 (cumulative)
For Blank cartridges only; Only one Perforation, not to exceed 2mm on butcher paper at 7 meters is allowed.	100-1-2	200-2-3 (cumulative)
For Blank cartridges only; Hangfire, Self-Firing, or Any cartridge that either causes injury to user, damages the weapon or renders it inoperable (without the use of special tools)	100-0-1	Critical, No Retest
Note 1: For order quantities less than 50,000 reference Appendix A for sample size.		

4.5.6 Bullet integrity. Bullet integrity tests shall be conducted during LAT in accordance with producer's test procedure to verify the sample meets the specified bullet integrity requirement.

4.5.7 Performance. Performance testing shall be conducted in accordance with the producer's test procedure to verify the sample meets the specified performance requirements (tracer, incendiary, blank, anti-armor) outlined in the TDP.

5. PACKAGING

5.1 Preservation, packaging, packing, unitization, and marking shall provide protection for multiple handling, redistribution, and shipment by any transportation mode and meet or exceed the following requirements.

5.1.1 Packaging containers for hazardous materials, ammunition and explosives shall meet or exceed the requirements found in part 6 of the "United Nations Recommendations on the Transport of Dangerous Goods Model Regulations" and in a manner acceptable to the competent authority of the nation of origin and in accordance with the regulations of all applicable carriers.

a. Type I Packaging (packaging for combat operations): Shall be robust packaging capable of withstanding off road/tactical transportation and a single exposure to unprotected outdoor storage conditions for not less than 6 months without degradation to the ammunition. Additionally, the packaging shall not require additional setup or modification to withstand outdoor storage once it leaves the manufacturer.

b. Type II Packaging (packaging for training use only): Commercial type packaging suitable for indoor storage; no additional requirements are implied.

5.1.2 Unit Package – Both Type I and Type II Unit package shall be so designed and constructed that it will contain the contents with no damage to the item(s), and with minimal damage to the unit pack during shipment and storage in the shipping container and will allow subsequent handling. The outermost component of the unit package shall be a container such as a sealed bag, carton, or box.

5.1.3 Cleanliness – Items and packaging shall be free of dirt and other contaminants which would contribute to the deterioration of the item, or which would require cleaning by the customer prior to use. Coatings and preservatives applied to the item for protection are not considered contaminants.

5.1.4 Preservation – Items susceptible to corrosion or deterioration shall be provided protection against external environmental effects.

5.1.5 Cushioning – Items requiring protection from physical and mechanical damage (example: fragile, sensitive, critical material) or which could cause physical damage to other items, shall be protected by wrapping, cushioning, pack compartmentalization, or other means to mitigate shock and vibration and prevent damage during handling and shipment.

5.1.6 Wood Packing Material (WPM) – All wood packing material shall be constructed from heat treated lumber and certified by an accredited recognized agency in accordance with International Standards for Phytosanitary Measures (ISPM-15) regardless of origin or destination. All natural wood packing materials must include certification markings that are readily visible to inspectors and on at least two sides of every container when palletized.

5.2. Packing

5.2.1 All shipping containers shall be the most cost effective and shall be of the minimum cube to contain and protect the items.

5.2.2 Shipping Containers – The shipping container (including any necessary blocking, bracing, cushioning, or waterproofing) shall comply with the regulations of the carrier used and shall provide safe delivery to the destination at the lowest tariff cost. The shipping container shall be capable of multiple handling, stacking at least ten feet high, and storage under favorable conditions and meet the requirements of the “United Nations Recommendations on the Transport of Dangerous Goods”.

5.3. Unitization

5.3.1 Shipments of identical items going to the same destination shall be palletized if they have a total cubic displacement of 20 cubic feet or more unless skids or other forklift handling features are included on the containers. Pallet loads must be stable, and to the greatest extent possible, provide a level top for ease of stacking. The weight capacity of the pallet must be adequate for the load. A pallet load shall not exceed 4,000 pounds and should not exceed 1321mm in length or width, or 1372mm in height. The load shall be contained in a manner that will permit safe handling during shipment and storage.

5.3.2 Banding – Metal banding shall be used to secure the load. Straps shall be applied to each column or layer of boxes. Tie down straps shall be applied to each column of boxes at 90 degrees to the load straps. Edge protectors shall be used when securing fiberboard boxes.

5.4. Marking

5.4.1 Packaging marking shall be visible, clear, and remain legible during normal life cycle handling.

5.4.2 All unit packages, intermediate packs, exterior shipping containers, and, as applicable, unitized loads shall be marked with item description, quantity, lot number, or serial number. The outer shipping container and unitized load shall indicate load weight, UN dangerous goods proper shipping name, and location of UN number and UN POP Marking.

5.4.3 Each package shipping container shall show the United Nations packaging symbol and applicable codes in accordance with the construction requirements and testing of packaging as expressed in part 6 of the "United Nations Recommendations on the Transport of Dangerous Goods".

5.5. Additional Requirements for Hazardous Materials

5.5.1 The shipment shall be prepared in accordance with the "United Nations Recommendations on the Transport of Dangerous Goods" and other applicable regulations effective at the time of shipment in a manner acceptable to the competent authority of the nation of origin and in accordance with the regulations of all applicable carriers.

5.5.2 Packaging and marking for hazardous material shall comply with the requirements for the mode of transport and the applicable performance packaging contained in the following documents:

- a. International air transport: International Civil Aviation Organization (ICAO) Technical Instructions for the Safe Transport of Dangerous Goods by Air.
- b. International vessel transport: International Maritime Organization (IMO) International.

6. REPORTING

6.1 Lot Acceptance Test Reports (LATR), at a minimum, must include a summary of findings for all visual and ballistic testing, lot history folders (in-process testing per TDP requirements, ammo data card, and all certifications or certificates), all measured data, pressure and velocity for each individual shot, range for all test shots, deflection for all test shots scored for dispersion, identify conditioning temperature used, calculated average pressure and velocity, spread for minimum and maximum velocity, and calculated dispersion. The LATR must also include the ammo data card for all lots tested.

6.1.1 Ammo Data Cards are required to be provided as part of the LATR and must include lot information for all energetics and major components used in the production of each lot.

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Annex A: Weapon Interface
Annex B: Operating Requirements
Annex C: Visual Inspection Requirements
APPENDIX - Definitions

ANNEX A – Weapon Interface

Small Caliber Ammunition Weapon Interface¹

Item	Weapon Interface¹
5.45 x 39mm Ball	AK-74 Assault Rifle
7.62 x 25mm Ball	Tokarev Pistol
7.62 x 39mm Ball	AK-47 Assault Rifle / RPK Light Machine Gun
7.62 x 39mm Tracer	AK-47 Assault Rifle / RPK Light Machine Gun
7.62 x 39mm Blank	AK-47 Assault Rifle / RPK Light Machine Gun
7.62 x 54mm R Ball	PKM Machine Gun / PKT Tank Machine Gun / SVD Dragunov Sniper Rifle
7.62 x 54mm R Tracer	PKM Machine Gun / PKT Tank Machine Gun
7.62 x 54mm R Armor Piercing	PKM Machine Gun / PKT Tank Machine Gun
7.62 x 54mm R Armor Piercing Incendiary	PKM Machine Gun / PKT Tank Machine Gun
7.62 x 54mm R Armor Piercing Tracer	PKM Machine Gun / PKT Tank Machine Gun
7.62 x 54mm R Blank	PKM Machine Gun / PKT Tank Machine Gun
7.62 x 54mm R 7N1 Sniper	SVD Dragunov Sniper/PSL Rifle
.338 Ball	Barrett MRAD/Rifle SAKO TRG 42
.338 Armor Piercing	Bellator Rifle
9 x 18mm Ball	Makarov Pistol
12.7 x 108mm Ball	YaKB Machine Gun / DshKM Machine Gun
12.7 x 108mm Tracer	YaKB Machine Gun / DshKM Machine Gun
12.7 x 108mm Armor Piercing	YaKB Machine Gun / DshKM Machine Gun
12.7 x 108mm Armor Piercing Incendiary	YaKB Machine Gun / DshKM Machine Gun
12.7 x 108mm Armor Piercing Tracer	YaKB Machine Gun / DshKM Machine Gun
12.7 x 108mm Armor Piercing Incendiary Tracer	YaKB Machine Gun / DshKM Machine Gun
12 Gauge 2¾” Rifled Slug	Shotgun M590 A1/M1014
Note 1: Similar type weapon may be used. The weapon shall be listed in TDP and must be approved for use by the USG.	

ANNEX A

Table 1: Function & Casualty Test for all Pistol, Shotgun, and Rifle up to 12.7mm

Test for order quantity 50,000+ cartridges	Temperature	Retest (Cumulative)
230-1-2	Hot	350-2-3
230-1-2	Ambient	350-2-3
230-1-2	Cold	350-2-3

Test for order quantity 5,000-49,999 cartridges	Temperature	Retest (Cumulative)
100-1-2	Hot	200-2-3
100-1-2	Ambient	200-2-3
100-1-2	Cold	200-2-3

Note 1: For orders less than 5000 cartridges function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

Table 2: Function & Casualty Test for Blank ammunition (all calibers)

Test for order quantity 5,000+ cartridges	Temperature	Retest (Cumulative)
100-2-3 ¹	Hot	200-4-5 ¹
100-2-3 ¹	Ambient	200-4-5 ¹
100-2-3 ¹	Cold	200-4-5 ¹

Test for order quantity 1000-4999 cartridges	Temperature	Retest (Cumulative)
100-2-3 ¹	Ambient	200-4-5 ¹

Note 1: The test requirements for perforation are 100-1-2 with a retest of 200-2-3.

Note 2: For orders less than 1000 cartridges function and casualty tests are not required. The USG reserves the right to increase these requirements in their proposals.

ANNEX B – Operating Requirements

Note: Any additional specifications not explicitly detailed in this document shall comply either with SAAMI or C.I.P. Standards. All technical parameters must be sourced from one single standard; specifications from different standards may not be mixed for any single item. When using the SAAMI or CIP, if any piece of data from the criteria in the tables below is missing, refer to the tables in Annex B.

9x18mm Technical Requirements						
Item ¹	Average Velocity (m/s) ²	Δ Velocity (m/s) ²	Max. Avg. chamber pressure (kgf/cm ²) ²	Individual max chamber pressure (kgf/cm ²) ²	Accuracy Mean Radius	
					At 25 meters	At 25 yards
9 x 18mm Ball, Lead Core	≥ 280	≤ 30	≤ 1700	N/A	≤ 4.8 cm	≤ 1.73 in
Note 1: Accuracy, Pressure, and Velocity testing shall have a minimum sample size of 10 test cartridges at ambient temperature. Specialized equipment such as accuracy barrels and ported fixtures are permitted for these tests. Multiple tests can be combined to reduce number of total cartridges fired: i.e. The same cartridge may be used to measure Pressure, Velocity and Accuracy provided the equipment is capable of taking all measurements at the same time. Note 2: The application of correction factors to pressure and velocity using results from a reference lot is permissible						

7.62 x 25mm Technical Requirements						
Item ¹	Average Velocity (m/s) ²	Δ Velocity (m/s) ²	Max. Avg. chamber pressure (kgf/cm ²) ²	Individual max chamber pressure (kgf/cm ²) ²	Accuracy Mean Radius	
					At 25 meters	At 25 yards
7.62 x 25mm Ball, Lead Core	≥ 475	≤ 25	≤ 2400	N/A	≤ 5 cm	≤ 1.8 in
Note 1: Accuracy, Pressure, and Velocity testing shall have a minimum sample size of 10 test cartridges at ambient temperature. Specialized equipment such as accuracy barrels and ported fixtures are permitted for these tests. Multiple tests can be combined to reduce number of total cartridges fired: i.e. The same cartridge may be used to measure Pressure, Velocity and Accuracy provided the equipment is capable of taking all measurements at the same time. Note 2: The application of correction factors to pressure and velocity using results from a reference lot is permissible.						

ANNEX B – Operating Requirements (continued)

.338 Technical Requirements								
Item¹	Bullet Weight (gr)	Average Velocity (fps)²	Δ Velocity (fps)²	Max Avg. chamber pressure (psi)²	Individual max chamber pressure (psi)²	Accuracy Extreme Spread		Penetration³
						At 100 yards ⁴	At 300 meters	At 100 yards
Lapua Ball, Lead Core	250	≥ 2,838	≤ 50	≤ 60,916	≤ 70,054	≤ 3.0 in	≤ 8.73 cm	N/A
Armor Piercing, Tungsten Carbide Core	300	≥ 2,560	≤ 90	≤ 62,500	≤ 64,100	≤ 3.0 in	≤ 3.0 in	≥ 0.375 in
Federal	210	≥ 2,520	≤ 90	≤ 62,000	≤ 63,600	≤ 2.5 in		N/A
Marlin Express	200	≥ 2,460	≤ 90	≤ 46,000	≤ 47,200	≤ 2.5 in		N/A
Remington Ultra Magnum	250	≥ 2,760	≤ 90	≤ 65,000	≤ 66,600	≤ 3.0 in		N/A
Ruger Compact Magnum	225	≥ 2,670	≤ 90	≤ 65,000	≤ 66,600	≤ 2.5 in		N/A
Weatherby Rebated Precision Magnum	225	≥ 2,710	≤ 90	≤ 65,000	≤ 66,600	≤ 2.5 in		N/A
Winchester Magnum	300	≥ 2,325	≤ 90	≤ 64,000	≤ 65,600	≤ 3.5 in		N/A
Note 1: Accuracy, Pressure, and Velocity testing shall have a minimum sample size of 10 test cartridges at ambient temperature. Specialized equipment such as accuracy barrels and ported fixtures is permitted for these tests. Multiple tests can be combined to reduce number of total cartridges fired: i.e. The same cartridge may be used to measure Pressure, Velocity and Accuracy provided the equipment is capable of taking all measurements at the same time. Note 2: The application of correction factors to pressure and velocity using results from a reference lot is permissible Note 3: Not less than 80% of the Armor piercing ammunition (10 shot minimum) shall have full penetration of armor. All rounds must be fired with no safety or critical failures. Note 4: Accuracy requirements from this column come from SAAMI and thus must follow SAAMI verification procedure firing 5 groups of 10 shots where each group must meet the accuracy requirement.								

ANNEX B – Operating Requirements (continued)

5.45x39mm Technical Requirements						
Item ¹	Average Velocity (m/s) ⁴	Δ Velocity (m/s) ⁴	Max. Avg. chamber pressure (kgf/cm ²) ⁴	Individual max chamber pressure (kgf/cm ²) ⁴	Accuracy Mean Radius	
					At 100 meters	At 100 yards
5.45 x 39mm Ball, Lead Core	≥ 785	≤ 30	≤ 3874	≤ 4456	≤ 10 cm	≤ 3.6 in
5.45 x 39mm Tracer ²	≥ 785	≤ 30	≤ 3874	≤ 4456	≤ 10 cm	≤ 3.6 in
5.45 x 39mm Armor Piercing ³	≥ 785	≤ 30	≤ 3874	≤ 4456	≤ 10 cm	≤ 3.6 in
Note 1: Accuracy, Pressure, and Velocity testing shall have a minimum sample size of 10 test cartridges at ambient temperature. Specialized equipment such as accuracy barrels and ported fixtures are permitted for these tests. Multiple tests can be combined to reduce number of total cartridges fired: i.e. The same cartridge may be used to measure Pressure, Velocity and Accuracy provided the equipment is capable of taking all measurements at the same time. Note 2: Not less than 80% of the Tracer cartridges (10 shot minimum) shall ignite and remain lit for a minimum time or distance as per the manufacturer's TDP. All tracers within the lot shall be of the same color. Note 3: Not less than 80% of the Armor piercing cartridges (10 shot minimum) shall have full penetration of armor with a thickness as per the manufacturer's TDP. Note 4: The application of correction factors to pressure and velocity using results from a reference lot is permissible						

ANNEX B – Operating Requirements (continued)

7.62x39mm Technical Requirements						
Item ¹	Average Velocity (m/s) ⁶	Δ Velocity (m/s) ⁶	Max. Avg. chamber pressure (kgf/cm ²) ⁶	Individual max chamber pressure (kgf/cm ²) ⁶	Accuracy Mean Radius	
					At 100 meters	At 100 yards
7.62 x 39mm Ball, Lead Core	≥ 710	≤ 35	$\leq 3620^4$ $\leq 3241^5$	$\leq 4160^4$ $\leq 3650^5$	≤ 20 cm	≤ 7.2 in
7.62 x 39mm Tracer ²	≥ 710	≤ 35	$\leq 3620^4$ $\leq 3241^5$	$\leq 4160^4$ $\leq 3650^5$	≤ 20 cm	≤ 7.2 in
7.62 x 39mm Armor Piercing ³	≥ 710	≤ 40	$\leq 3620^4$ $\leq 3241^5$	$\leq 4160^4$ $\leq 3650^5$	≤ 20 cm	≤ 7.2 in
Note 1: Accuracy, Pressure, and Velocity testing shall have a minimum sample size of 10 test cartridges at ambient temperature. Specialized equipment such as accuracy barrels and ported fixtures are permitted for these tests. Multiple tests can be combined to reduce number of total cartridges fired: i.e. The same cartridge may be used to measure Pressure, Velocity and Accuracy provided the equipment is capable of taking all measurements at the same time. Note 2: Not less than 80% of the Tracer cartridges (10 shot minimum) shall ignite and remain lit for a minimum time or distance as per the manufacturer's TDP. All tracers within the lot shall be of the same color. Note 3: Not less than 80% of the Armor piercing cartridges (10 shot minimum) shall have full penetration of armor with a thickness as per the manufacturer's TDP. Note 4: Pressure at ambient temperature determined using copper crusher. Note 5: Pressure at ambient temperature determined using piezoelectric transducer. Note 6: The application of correction factors to pressure and velocity using results from a reference lot is permissible						

ANNEX B – Operating Requirements (continued)

7.62x54mm R Technical Requirements						
Item ¹	Average Velocity (m/s) ⁵	Δ Velocity (m/s) ⁵	Max. Avg. chamber pressure (kgf/cm ²) ⁵	Individual max chamber pressure (kgf/cm ²) ⁵	Accuracy Mean Radius	
					At 100 meters	At 100 yards
7.62 x 54mm R Ball, Lead Core	≥ 780	≤ 40	≤ 3975	≤ 4570	≤ 15 cm	≤ 5.4 in
7.62 x 54mm R Tracer ²	≥ 780	≤ 40	≤ 3975	≤ 4570	≤ 15 cm	≤ 5.4 in
7.62 x 54mm R Armor Piercing ³	≥ 780	≤ 40	≤ 3975	≤ 4570	≤ 15 cm	≤ 5.4 in
7.62 x 54mm R Armor Piercing ³ Tracer ²	≥ 780	≤ 40	≤ 3975	≤ 4570	≤ 15 cm	≤ 5.4 in
7.62 x 54mm R Armor Piercing ³ Incendiary ⁴	≥ 780	≤ 40	≤ 3975	≤ 4570	≤ 15 cm	≤ 5.4 in
7.62 x 54mm R 7N1 Sniper	≥ 775	≤ 25	≤ 3975	≤ 4570	≤ 2.0 cm	≤ 0.72 in
Note 1: Accuracy, Pressure, and Velocity testing shall have a minimum sample size of 10 test cartridges at ambient temperature. Specialized equipment such as accuracy barrels and ported fixtures are permitted for these tests. Multiple tests can be combined to reduce number of total cartridges fired: i.e. The same cartridge may be used to measure Pressure, Velocity and Accuracy provided the equipment is capable of taking all measurements at the same time. Note 2: Not less than 80% of the Tracer cartridges (10 shot minimum) shall ignite and remain lit for a minimum time or distance as per the manufacturer's TDP. All tracers within the lot shall be of the same color. Note 3: Not less than 80% of the Armor piercing cartridges (10 shot minimum) shall have full penetration of armor with a thickness as per the manufacturer's TDP. Note 4: Not less than 80% of the Incendiary cartridges (10 shot minimum) shall, upon impact, develop an incandescent flash sufficiently hot, intense and sustained to ignite explosive vapors or initiate combustion in readily ignitable materials. Note 5: The application of correction factors to pressure and velocity using results from a reference lot is permissible						

ANNEX B – Operating Requirements (continued)

12.7x108mm Technical Requirements								
Item ¹	Average Velocity (m/s) ⁵	Δ Velocity (m/s) ⁵	Max. Avg. chamber pressure (kgf/cm ²) ⁵	Individual max chamber pressure (kgf/cm ²) ⁵	Accuracy Mean Radius			
					At 300 meters	At 300 yards	At 100 meters	At 100 yards
12.7 x 108mm Ball; Lead Core	≥ 810	≤ 40	≤ 3175	≤ 3650	≤ 30 cm	≤ 10.8 in	≤ 10 cm	≤ 3.6 in
12.7 x 108mm Tracer ²	≥ 810	≤ 40	≤ 3175	≤ 3650	≤ 30 cm	≤ 10.8 in	≤ 10 cm	≤ 3.6 in
12.7 x 108mm Armor Piercing ³	≥ 810	≤ 40	≤ 3175	≤ 3650	≤ 30 cm	≤ 10.8 in	≤ 10 cm	≤ 3.6 in
12.7 x 108mm Armor Piercing ³ Tracer ²	≥ 810	≤ 40	≤ 3175	≤ 3650	≤ 30 cm	≤ 10.8 in	≤ 10 cm	≤ 3.6 in
12.7 x 108mm Armor Piercing ³ Incendiary ⁴	≥ 810	≤ 40	≤ 3175	≤ 3650	≤ 30 cm	≤ 10.8 in	≤ 10 cm	≤ 3.6 in
12.7 x 108mm Armor Piercing ³ Incendiary ⁴ Tracer ²	≥ 810	≤ 40	≤ 3175	≤ 3650	≤ 30 cm	≤ 10.8 in	≤ 10 cm	≤ 3.6 in
Note 1: Accuracy, Pressure, and Velocity testing shall have a minimum sample size of 10 test cartridges at ambient temperature. Specialized equipment such as accuracy barrels and ported fixtures are permitted for these tests. Multiple tests can be combined to reduce number of total cartridges fired: i.e. The same cartridge may be used to measure Pressure, Velocity and Accuracy provided the equipment is capable of taking all measurements at the same time. Note 2: Not less than 80% of the Tracer cartridges (10 shot minimum) shall ignite and remain lit for a minimum time or distance as per the manufacturer's TDP. All tracers within the lot shall be of the same color. Note 3: Not less than 80% of the Armor piercing cartridges (10 shot minimum) shall have full penetration of armor with a thickness as per the manufacturer's TDP. Note 4: Not less than 80% of the Incendiary cartridges (10 shot minimum) shall, upon impact, develop an incandescent flash sufficiently hot, intense and sustained to ignite explosive vapors or initiate combustion in readily ignitable materials. Note 5: The application of correction factors to pressure and velocity using results from a reference lot is permissible								

ANNEX C – Test Requirements and Classification

TABLE III. Small Caliber Ammunition Visual Inspection Requirements

Critical: 150-0-1	Major: 150-5-6 Reinspection: 450-15-16 (cumulative)	Minor: 150-9-10 Reinspection: 450-27-28 (cumulative)
Mixed ammunition types. (Combat cartridge mixed with Blank cartridge lot)	Minor surface corrosion	Discolored, oily, dirty, or smeared
Spillage/leakage of energetic materials	Cracks or perforations present; not allowing propellant spillage or considered critical	Head stamp missing, illegible or does not match others in lot
Case cracks or perforations present; allowing propellant spillage	Mixed ammunition types (Blank cartridge mixed with Combat cartridge lot)	Bullet tip Color identifier(s) present but improperly applied
Splits in the case around primer area, case groove or case rim	Inverted or missing bullet	Missing or incomplete sealant (if applicable)
Gross deformation and/or gross corrosion	Bullet tip Color identifier(s) missing (if applicable)	Dents, Folds, or Wrinkles on case
Bourrelet diameter too large	Primer missing, inverted, double, cocked, or visually above flush	Damaged bullet point
	Total cartridge length out of spec ^{4,5}	
	Incorrect quantity of cartridges in packaging	
Note 1: For an order quantity greater than 5000 cartridges a minimum of 150 cartridges must be visually inspected per lot. Note 2: If the number of allowable defects is exceeded a double sample quantity of 300 cartridges shall be re-inspected. The accept/reject criteria is based on the cumulative number of defects for all three samples. Note 3: If a critical is found or the number of allowable majors or minors, after reinspection, is exceeded the lot fails. Note 4: Cartridges that measure less than the minimum overall length shall be set aside for Ballistic Pressure Testing. If pressure requirements are not met, the defect counts as a critical. Note 5: Cartridges that measure more than the maximum overall length shall be set aside for Function Testing. All function requirements must be met. Note 6: The inspection and reinspection quantity listed is selected during LAT. DCMA will utilize their Zero-Based Acceptance Sample Plans shown in Appendix during GSI.		

APPENDIX

1. DEFINITIONS

1.1 PdL SAWS Small Caliber Ammunition. PdL SAWS small caliber ammunition are those munitions that have not been safety tested and type classified for Army use, and cannot be procured through the Army supply. Munitions and explosives that are not managed by National Inventory Control Points, have not been safety tested nor type classified for Army use, do not have a national stock number (NSN) and cannot be procured or requisitioned through the Army or other Department of Defense supply system. Projectile diameter shall be 13.95mm or less.

1.2 Technical data. Technical data, otherwise known as the Technical Data Package (TDP), is the product specific technical drawings and Quality Assurance (QA) requirements to which ammunition and associated packaging are produced and accepted for each applicable Contract Line Item Number (CLIN). For the purpose of this specification, technical data, as a minimum, must contain the following: The Government Contract Number, Delivery Order, CLIN, top assembly drawings with revision number, revision date, interface dimensions, and list of component assemblies and drawings with revision dates and key assembly dimensions; packaging and marking drawings with revision number, revision date, and markings; product specification with visual, dimensional and ballistic acceptance test methods, including in-process testing, with sample size, accept/reject criteria, and retest criteria. The cover page of the TDP must indicate if it is for new production or stock ammunition.

1.3 Degradation. Ammunition with gross nonconformance to identified technical requirements, corrosion, cracks, deformation and spillage.

1.4 Deterioration. Packaging that is ripped, broken, perforated, water damaged, or crushed.

1.5 Skip-Lot Temperature Testing. Testing at the operational temperature limits shall be conducted on the first lot of every delivery order and every 5th lot thereafter (1, 5, 10, 15...). The ammunition must pass the requirements defined in ANNEX A. If a lot is rejected for any reason, the following two lots must pass extreme temperature testing before resuming the skip lot procedures.

APPENDIX (continued)

	Acceptable Quality Level (AQL)															
LOT SIZE	0.01	0.015	0.025	0.04	0.065	0.10	0.15	0.25	0.40	0.65	1.0	1.5	2.5	4.0	6.5	10.0
1-8	A	A	A	A	A	A	A	A	A	A	A	A	5	3	3	3
9-15	A	A	A	A	A	A	A	A	A	A	13	8	5	3	3	3
16-25	A	A	A	A	A	A	A	A	A	20	13	8	5	3	3	3
26-50	A	A	A	A	A	A	A	A	32	20	13	8	7	7	5	3
51-90	A	A	A	A	A	A	80	50	32	20	13	13	11	8	5	4
91-150	A	A	A	A	A	125	80	50	32	20	19	19	11	9	6	5
151-280	A	A	A	A	200	125	80	50	32	29	29	19	13	10	7	6
281-500	A	A	A	315	200	125	80	50	48	47	29	21	16	11	9	7
501-1200	A	800	500	315	200	125	80	75	73	47	34	27	19	15	11	8
1201-3200	1250	800	500	315	200	125	120	116	73	53	42	35	23	18	13	9
3201-10,000	1250	800	500	315	200	192	189	116	86	68	50	38	29	22	15	9
10,001-35,000	1250	800	500	315	300	294	189	135	108	77	60	46	35	29	15	9
35,001-150,000	1250	800	500	490	476	294	218	170	123	96	74	56	40	29	15	9
150,001-500,000	1250	800	750	715	476	345	270	200	156	119	90	64	40	29	15	9
500,001 & Over	1250	1200	1112	715	556	435	303	244	189	143	102	64	40	29	15	9

Note: An "A" in place of a number indicates that all units in a lot must be inspected